THE FOUR LEMMA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the first assertion, with both applications of the snake lemma written out; the closing sentence 'The proof of (2) is dual to this.' is the author's own routine duality remark on the part written out in full inside the counted span, and it is preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The snake lemma. Same chapter, Lemma lemma-snake. Supplies the connecting map and the exactness of the associated six-term sequence, applied twice inside the proof to extract the two surjectivity statements.

Source reference. Original lemma 010H: sources/stacks-homology/pinned-original-source.tex, source lines 1229–1229. Navigation pointer invoked at proof line 1421; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccccccc} w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} \\ w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

be a commutative diagram with exact rows.

(1) *If α, γ are surjective and δ is injective, then β is surjective.*
(2) *If β, δ are injective and α is surjective, then γ is injective.*

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Proof. Assume α, γ are surjective and δ is injective. We may replace w' by $\mathrm{Im}(w' \to x')$, i.e., we may assume that $w' \to x'$ is injective. We may replace z by $\mathrm{Im}(y \to z)$, i.e., we may assume that $y \to z$ is surjective. Then we may apply Lemma 010H to

$$\begin{array}{ccccccc} & & \mathrm{Ker}(y \to z) & \longrightarrow & y & \longrightarrow & z & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Ker}(y' \to z') & \longrightarrow & y' & \longrightarrow & z' & & \end{array}$$

to conclude that $\mathrm{Ker}(y \to z) \to \mathrm{Ker}(y' \to z')$ is surjective. Finally, we apply Lemma 010H to

$$\begin{array}{ccccccc} & & w & \longrightarrow & x & \longrightarrow & \mathrm{Ker}(y \to z) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & w' & \longrightarrow & x' & \longrightarrow & \mathrm{Ker}(y' \to z') & & \end{array}$$

to conclude that $x \to x'$ is surjective. This proves (1). The proof of (2) is dual to this. $\square$